Midterm Exam

Course Title	General physics lab.1	Course Code	30201111
Semester	Second	Academic Year	2023-2024
Exam Date	8-4-2024	Day	Monday
Time	13:30-14:20		

Student Name	[redacted]	Student ID No.		Section	[redacted]

Note: When you need consider the acceleration of gravity (g) =10 m/s², π=3.14

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. Student's signature: ------------

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Multiple Choice	CLIO, 1,2,3,8,9,10	20(66.7%)	14		
2	Essay	CLIO, 1,2,3,8,9,10	10(33.3%)	8		
Total			30(100%)	20		22

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

Fill your answers in the following table:

Q.1	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	D	A	B	C	D	A	C

Q.2	1	2	3	4	5
Ans.	0.6.1mm	73.1mm	4r....., 93.7	[illegible]	6.2

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Faculty of Engineering Technology
Department of Scientific Basic Sciences

قسم العلوم الأساسية العلمية

Midterm Exam Physics -111

Time of exam: 50 minutes

Q1: Circle the correct answers, and then move these answers to the table: (20marks)

1) Perform the following calculation: $[\frac{1}{2}(9.80)(0.5^2)]$, round your final answer according to the rules of the significant figures?

A. 1.225 B. 1.2 C. 1.0 [✓ D. 1]

2) One of the following readings measured by a Vernier caliper instrument:

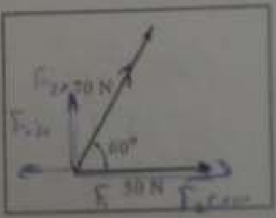
A. 2.1 cm [✓ B. 12.35 mm] C. 8.28 cm D. 1.823 cm

3) The micrometer is more accurate than the Vernier Caliper.

[✓ A. True] B. False

4) Find the direction of the resultant force with the positive x-axis of the two forces given in the figure below.

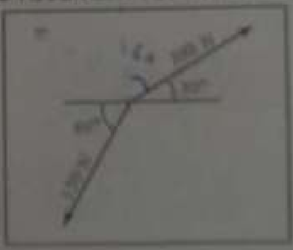
- A. 104.4°
- B. 35.5°
- C. 30.0°
- [✓ D. 54.5°]



Fy = 60.6
Fx = 85

5) Using cosine law to find the magnitude of the resultant force for the system of forces in the figure below:

- [✓ A. 80.74 N]
- B. 241.82 N
- C. 260.88 N
- D. 30.34 N



6) A rectangular plate has a length of (21.3 ± 0.2) cm and a width of (9.80 ± 0.10) cm. Calculate the area of the plate, including its uncertainty (A ± ΔA) (in cm²).

A. 208 ± 4 [✓ B. 209 ± 4] C. 208.7 ± 4 D. 208.0 ± 0.3

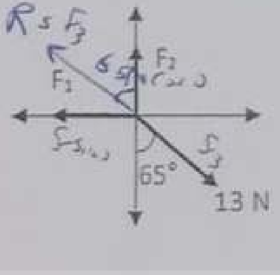
A = L × w

7) A carpet is to be installed in a rectangular room whose length is measured to be 12.71 m and whose width is measured to be 3.46 m. Find the area (in m²) of the room? (using S.F rules)

A. 43.98 B. 43.9 [✓ C. 44.0] D. 43.99

8) If you had measured the mass of a metallic sphere in the physics laboratory by digital device as ($m = 75.5 \pm 0.1\ gram$), and the true value is 80 gram, then the percentage error:

A. 0.001% B. 0.05% C. 0.005% [✓ D. 5.6%]

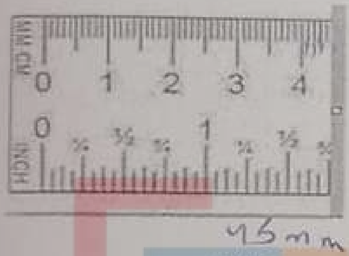


9) Three forces are in equilibrium, the magnitude of (F₁) is

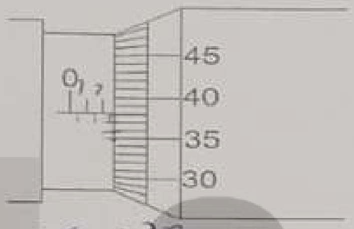
[✓ A. 11.8 N] B. -11.8 N C. 5.5N D. -5.5 N

55

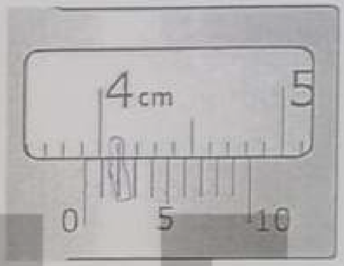
10) Using significant figures rules, find the sum of these readings:



45 mm
44.5



2.5 + 0.38
2.88 mm



39 mm + 0.2
39.2
86.6

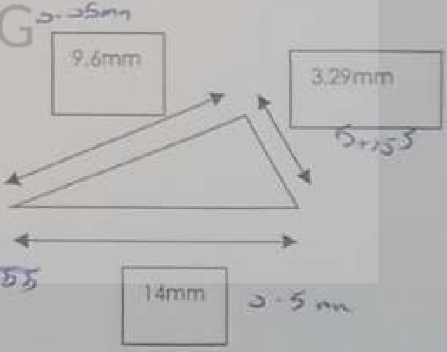
A. 86.mm B. 86.08 mm [✓ C. 86.1 mm] D. 46.5mm

Q.2) Fill in the following blanks with the correct answer: (10 Marks)

1) Student used different instruments to calculate the circumference of triangle, the error in the circumference is (using (S.F) rules is ....~~ΔC~~...

ΔC = 1mm

C = 27



0.05mm

0.005

0.5 mm

ΔC = 1

2) Micrometer reading is (8.74 mm), if we turned it (8) turns and (2/5) of turn, towards closing direction, its reading will be.....23.1.mm

$5 \times (5 - \frac{2}{5}$

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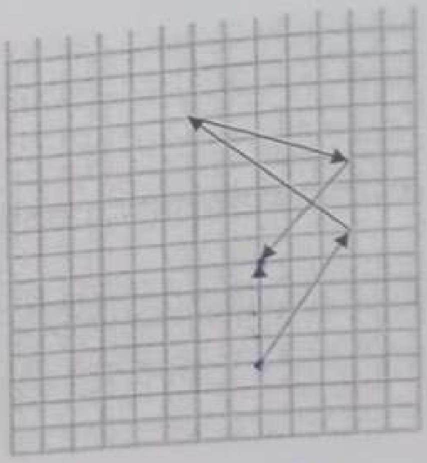
3) In the vector diagram shown, ((as each square (1 N))).

Find the magnitude and direction of resultant.

The magnitude is ....4 N........and the direction is ....90°..............



1 cm

4) A cylinder has a radius of R ± ΔR = (2.00 ± 0.02) cm, m ± Δm = (18.20 ± 0.01) g and a length of L ± ΔL= (10.00 ± 0.01) cm.

The error in the volume (ΔV) =.......3...................... cm³.

$V = \pi r^2 L =$ 126

~~$\Delta V = 2\pi r \Delta r$~~

~~$1\pi V$~~ $\Delta V = 2.646$

5) Three forces are in equilibrium depending on the shown table,

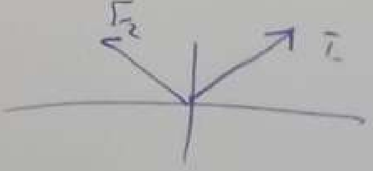
the magnitude of (F_3) is:

F_3 = ..6.2......N.

$F_1 = 3.6$

$F_2 = 5.7$

F	Θ	F_X	F_Y
F_1	30°	3.118	1.8
F_2	130°	-3.664	4.366
F_3	....	....	....



GOOD LUCK